Applied Machine Learning



Feature Selection
Introduction

Learning goals

- Reasons for Feature Selection
- Types of Feature Selection



Recap

WHAT IS FEATURE ENGINEERING?



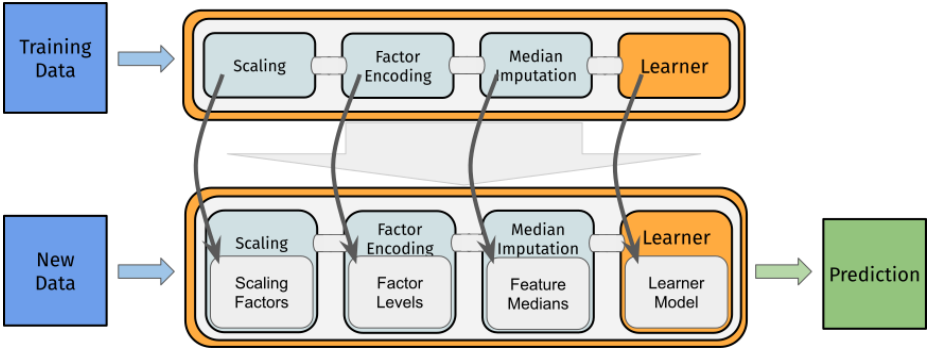
Feature engineering is on the intersection of **data cleaning**, **feature creation** and **feature selection**.

The goal is to solve common difficulties in data science projects, like

- skewed/weird feature distributions,
- (high cardinality) categorical features,
- functional (temporal) features,
- missing observations,
- high dimensional data,
- ...

and improve model performance (or make the data readable by the model in the first place).

TRAIN-TEST LEAKAGE



TYPES OF FEATURE ENGINEERING (SO FAR)



- numerical encoding of categorical features
- feature transformations
- imputation

Feature Selection

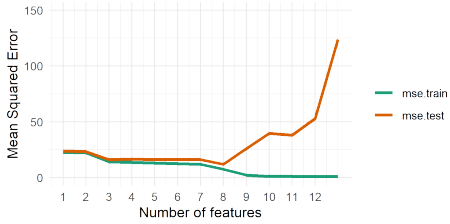


MOTIVATION I



- Naive view:
  - More features $\rightarrow$ more information $\rightarrow$ discriminant power $\uparrow$
  - Model is not harmed by irrelevant features since their parameters can simply be estimated as 0.
- In practice, irrelevant and redundant features can "confuse" learners (see **curse of dimensionality**) and worsen performance.
- Example: In linear regression, R^2 is monotonically increasing in p, but adding irrelevant features leads to overfitting (capturing noise).

MOTIVATION II





MOTIVATION III



- In high-dimensional data sets, we often have prior information that many features are either irrelevant or of low quality
- Having redundant features can cost something during prediction (money or time)
- Many models require $n > p$ data. Thus, we either need to
  - adapt models to high-dimensional data (e.g., regularization)
  - design entirely new procedures for $p > n$ data
  - use filter preprocessing methods from this lecture

SIZE OF DATASETS I



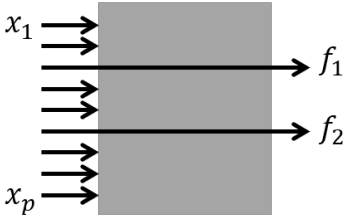
Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

- **Classical setting**: Up to around 10^2 features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality**: At around 10^2 to 10^3 features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data**: 10^3 to 10^9 or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called $p \gg n$.

FEATURE SELECTION VS. EXTRACTION I

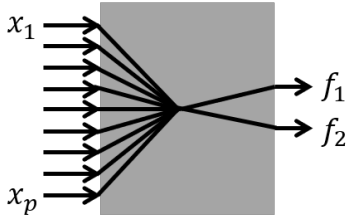


Feature selection



- Creates a subset of original features $\boldsymbol{x}$ by selecting $\tilde{p} < p$ features $\boldsymbol{f}$.
- Retains information on selected individual features.

Feature extraction



- Maps p features in $\boldsymbol{x}$ to $\tilde{p}$ extracted features $\boldsymbol{f}$.
- Info on individual features can be lost through (non-)linear combination.

REASONS FOR FEATURE SELECTION



- improved predictive performance, since we reduce overfitting on irrelevant features,
- robust models that do not rely on noisy features,
- simpler models that are easier to interpret,
- faster model fitting, e.g. for model updates,
- faster prediction, and
- no need to collect potentially expensive features.

GOALS OF FEATURE SELECTION



Single Feature Selection Problem: identifying a minimal-size subset of the variables that is optimally predictive for an outcome variable T of interest.

Multiple Feature Selection Problem: Let $\mathcal{S}$ be the solution to the single feature selection problem (called the reference solution). The solution $\mathcal{M}$ to the multiple feature selection problem consists of all minimal-size sets $\mathcal{S}_i \in \mathcal{F}$ that are statistically equivalent to $\mathcal{S}$.

- Important for knowledge discovery (know all possible solutions)
- We will not delve into this any further

TYPES OF FEATURE SELECTION METHODS I



In rest of the chapter, we introduce different types of methods for FS:

- Filters: evaluate relevance of features using statistical properties such as correlation with target variable
- Wrappers: use a model to evaluate subsets of features
- Embedded methods: integrate FS directly into specific model - we look at them in their dedicated chapters (e.g., CART, L_0, L_1)

Example: embedded method (Lasso) regularizing model params with $L1$ penalty enables “automatic" feature selection:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta}\|_1 = \sum_{i=1}^{n} \left(y^{(i)} - \boldsymbol{\theta}^\top \boldsymbol{x}\right)^2 + \lambda \sum_{j=1}^{p} |\theta_j|$$

TYPES OF FEATURE SELECTION METHODS II

